ChIP-seq

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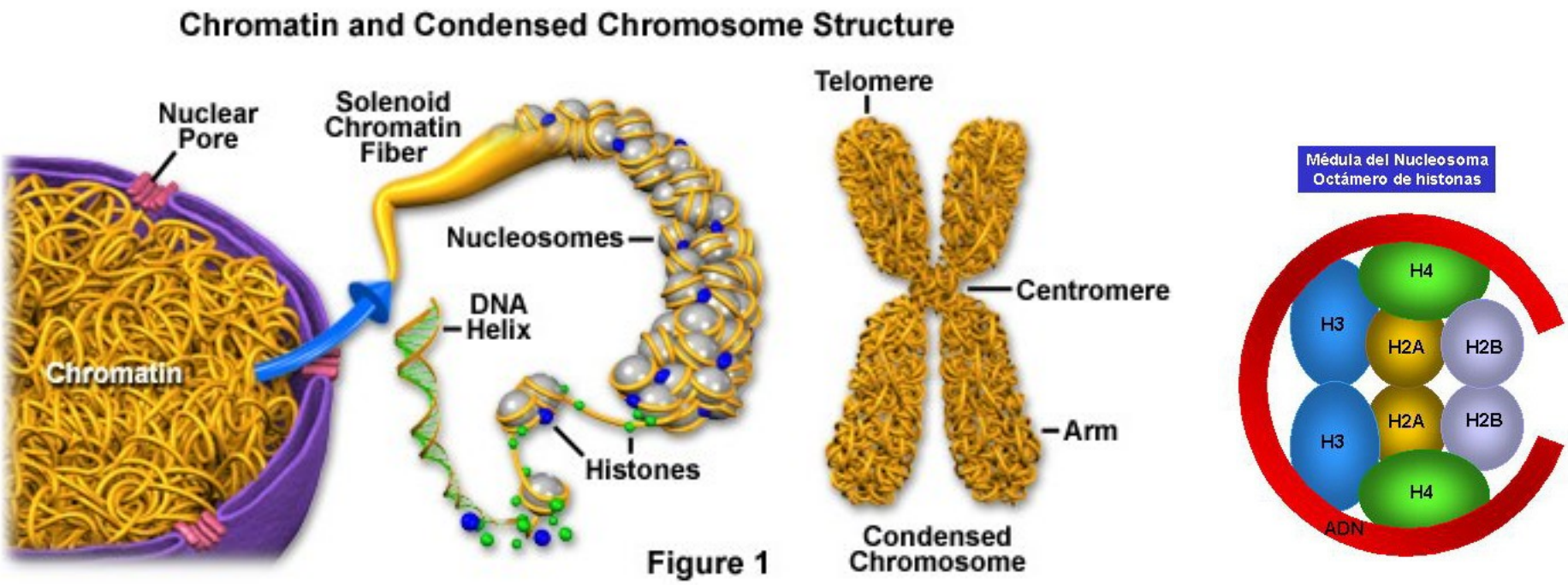
Chromatine, chromosomes, DNA



ChIP-seq

Chromatin immunoprecipitation followed by sequencing (ChIP-seq), allows in vivo determination of where a protein binds the genome, which can be transcription factors, DNA-binding enzymes, histones, chaperones, or nucleosomes.

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PLOS | COMPUTATIONAL BIOLOGY

Education

Practical Guidelines for the Comprehensive Analysis of ChIP-seq Data

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Summary: Mapping the chromosomal locations of transcription factors, nucleosomes, histone modifications, chromatin remodeling enzymes, chaperones, and polymerases is one of the key tasks of modern biology, as evidenced by the Encyclopedia of DNA Elements (ENCODE) Project. To this end, chromatin immunoprecipitation followed by high-throughput sequencing (ChIP-seq) is the standard methodology. Mapping such protein-DNA interactions *in vivo* using ChIP-seq presents multiple challenges not only in sample preparation and sequencing but also for computational analysis. Here, we present step-by-step guidelines for the computational analysis of ChIP-seq data.

where a protein binds the genome, which can be transcription factors, DNA-binding enzymes, histones, chaperones, or nucleosomes. ChIP-seq first cross-links bound proteins to chromatin, fragments the chromatin, captures the DNA fragments bound to one protein using an antibody specific to it, and sequences the ends of the captured fragments using next-generation sequencing (NGS). Computational mapping of the sequenced DNA identifies the genomic locations of bound DNA-binding enzymes, modified histones, chaperones, nucleosomes, and transcription factors (TFs), thereby illuminating the role of these protein-DNA interactions in gene expression and other cellular processes. The use of NGS provides relatively high resolution, low noise, and high genomic coverage

genome and the number and size of the binding sites of the protein. For mammalian transcription factors (TFs) and chromatin modifications such as enhancer-associated histone marks, which are typically localized at specific, narrow sites and have on the order of thousands of binding sites, 20 million reads may be adequate (4 million reads for worm and fly TFs) [7]. Proteins with more binding sites (e.g., RNA Pol II) or broader factors, including most histone marks, will require more reads, up to 60 million for mammalian ChIP-seq [8]. Importantly, control samples should be sequenced significantly deeper than the ChIP ones in a TF experiment and in experiments involving diffused broad-domain chromatin data. This is to ensure sufficient coverage of a substantial portion

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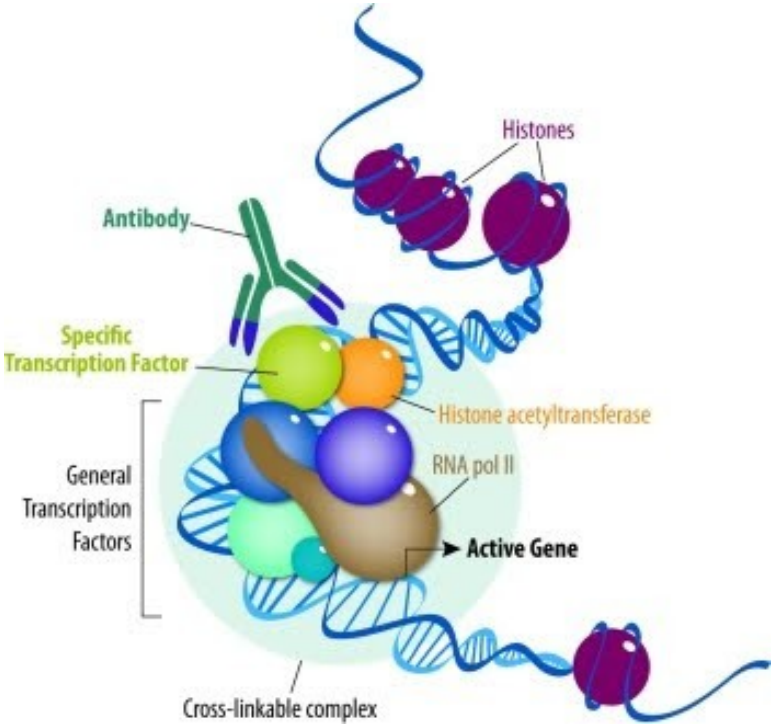
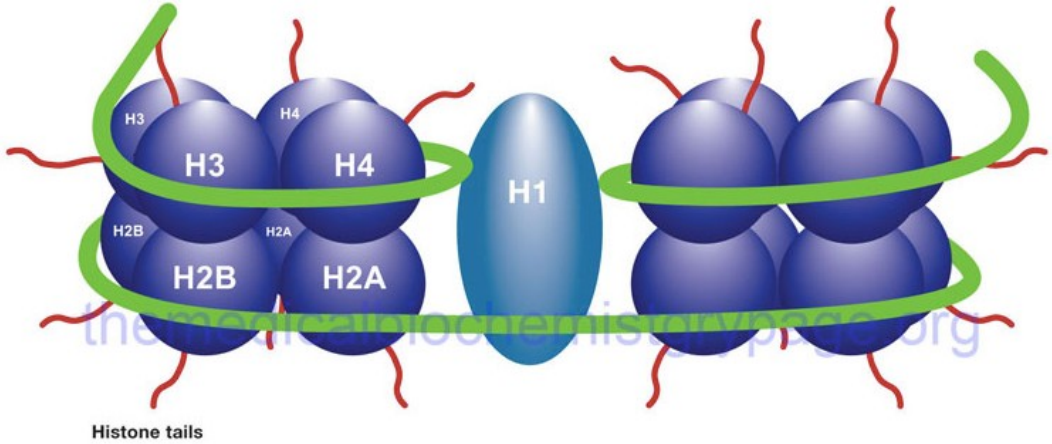
ChIP-seq


Histones
Antibody
Specific
Transcription Factor
Histone acetyltransferase
RNA pol II
General
Transcription
Factors
Active Gene
Cross-linkable complex


H3
H4
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Histone tails

REVIEWS

APPLICATIONS OF NEXT-GENERATION SEQUENCING

ChIP–seq: advantages and challenges of a maturing technology

Peter J. Park

Abstract | Chromatin immunoprecipitation followed by sequencing (ChIP–seq) is a technique for genome-wide profiling of DNA-binding proteins, histone modifications or nucleosomes. Owing to the tremendous progress in next-generation sequencing technology, ChIP–seq offers higher resolution, less noise and greater coverage than its array-based predecessor ChIP–chip. With the decreasing cost of sequencing, ChIP–seq has become an indispensable tool for studying gene regulation and epigenetic mechanisms. In this Review, I describe the benefits and challenges in harnessing this technique with an emphasis on issues related to experimental design and data analysis. ChIP–seq experiments generate large quantities of data, and effective computational analysis will be crucial for uncovering biological mechanisms.

Nucleosome
The basic structural subunit of chromatin. A nucleosome consists of approximately 147 base pairs of DNA and an octamer of histone proteins.

Epigenome
The chromatin states that are found along the genome, defined for a given time point and cell type. Thus, for a given genome there may be hundreds or thousands of epigenomes, depending on the stability of the chromatin states.

DNase I hypersensitive site
A chromosomal region that is highly accessible to cleavage by DNase I. Such sites are associated with open chromatin conformations and transcriptional activity.

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Genome-wide mapping of protein–DNA interactions and epigenetic marks is essential for a full understanding of transcriptional regulation. A precise map of binding sites for transcription factors, core transcriptional machinery and other DNA-binding proteins is vital for deciphering the gene regulatory networks that underlie various biological processes[1]. The combination of nucleosome positioning and dynamic modification of DNA and histones has a key role in gene regulation[2–4] and guides development and differentiation[5]. Chromatin states can influence transcription directly by altering the packaging of DNA to allow or prevent access to DNA-binding proteins, or they can modify the nucleosome surface to enhance or impede recruitment of effector protein complexes. Recent advances suggest that this interplay between chromatin and transcription is dynamic and more complex than previously appreciated[6], and there is a growing recognition that systematic profiling of the epigenomes in multiple cell types and stages may be needed for understanding developmental processes and disease states[7].

The main tool for investigating these mechanisms is chromatin immunoprecipitation (ChIP), which is a technique for assaying protein–DNA binding *in vivo*[8]. In ChIP, antibodies are used to select specific proteins or nucleosomes, which enriches for DNA fragments that are bound to these proteins or nucleosomes. The introduction of microarrays allowed the fragments obtained from ChIP to be identified by hybridization to a microarray (ChIP–chip), therefore enabling a genome-scale view of DNA–protein interactions[9,10]. On high-density tiling arrays, oligonucleotide probes can now be placed across an entire genome or across selected regions of a genome — for instance, promoter regions, specific chromosomes or gene families — at a preferred resolution.

Owing to the rapid technological developments in next-generation sequencing (NGS), the arsenal of genomic assays available to the biologist has been transformed[11–13]. The ability to sequence tens or hundreds of millions of short DNA fragments in a single run is enabling increasingly large experiments that could only be imagined a few years ago. Next-generation sequencing has already been applied in many areas, including the sequencing of whole genomes[14,15], the sequencing of mRNA for gene expression profiling (RNA–seq)[16–18], the characterization of structural variation[19], the profiling of DNase I hypersensitive sites[20], the detection of fusion genes from mRNA transcripts[21] and the discovery of new classes of small RNAs[22]. If the 'third-generation' sequencing technologies that are under development deliver as promised, they will lead to another epoch of genome-scale investigations[23].

Chromatin immunoprecipitation followed by sequencing (ChIP–seq) was one of the early applications of NGS, and the first studies to use it were published in 2007 (REFS 24–27). In ChIP–seq, the DNA fragments of interest are sequenced directly instead of being hybridized on an array. ChIP–seq has higher resolution, fewer artefacts, greater coverage and a larger dynamic range

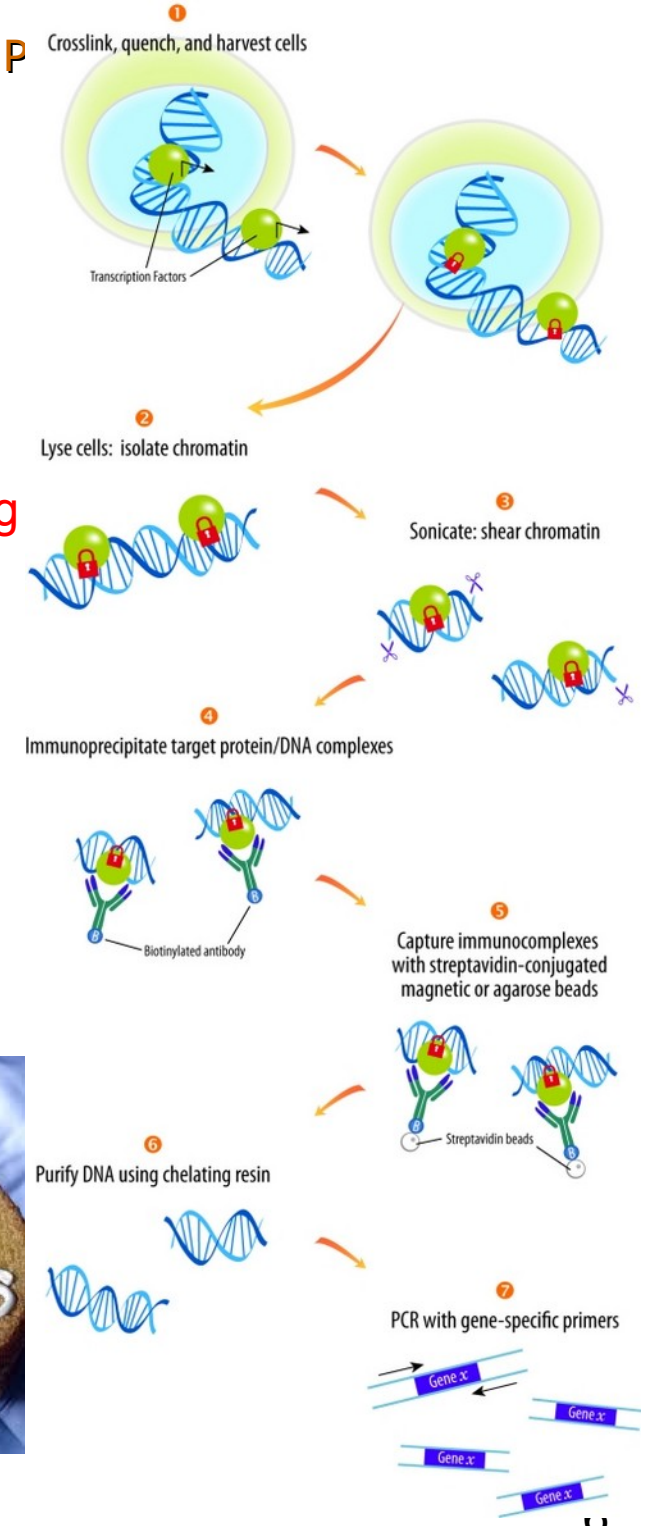
Nat Rev Genet. 2009 Oct;10(10):669-80.

ChIP-seq basics

In a ChIP experiment for DNA-binding proteins,
DNA fragments associated with a specific protein are enriched.

(1)The DNA-binding protein is crosslinked to DNA *in vivo* by treating cells with formaldehyde and (2)the chromatin is sheared by sonication into small fragments.

(3)An antibody specific to the protein of interest is used to immunoprecipitate the DNA-protein complex. Finally, (4)the crosslinks are reversed and the released DNA is assayed to determine the sequences bound by the protein.



Size selection
During the construction of a sequencing library, the immunoprecipitated DNA is subjected to size selection (typically in the ~150-300bp range).

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ChIP-seq, mapping

Computational mapping of the sequenced DNA identifies genomic locations of bound DNA-binding enzymes, modified histones, chaperones, nucleosomes, and transcription factors, thereby illuminating the roles of these protein-DNA interactions in gene expresion and other cellular processes.

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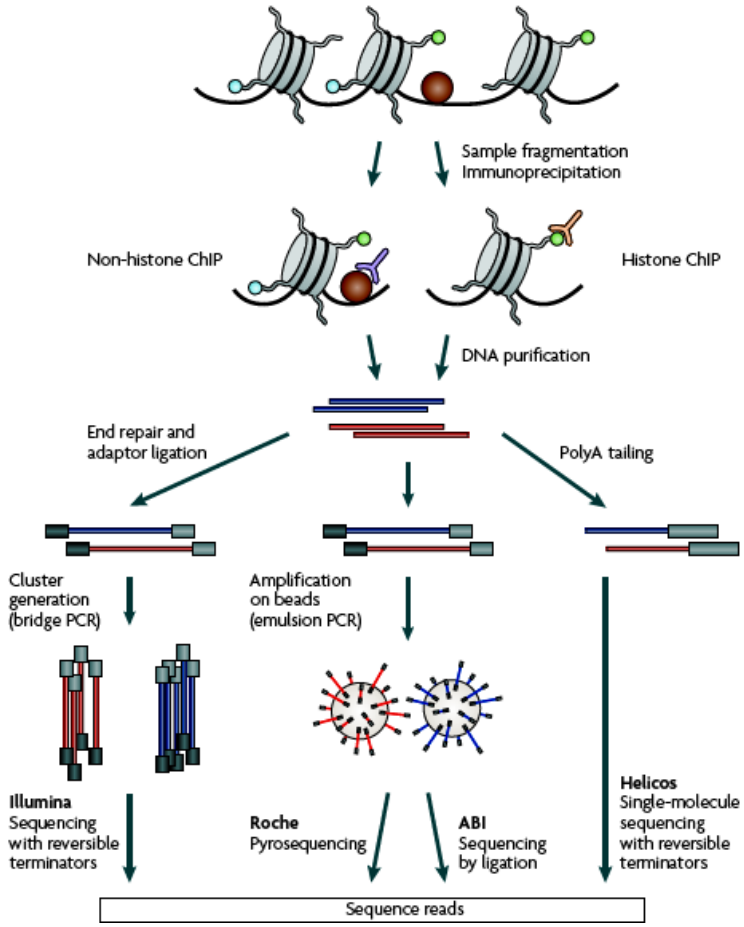

ChIP-seq, why?

Genome-wide mapping of protein-DNA interactions and epigenetic marks is essential for a full understanding of transcriptional regulation.

A precise map of binding sites for transcription factors, core transcriptional machinery and other DNA-binding proteins is vital for deciphering the gene regulatory networks that underlie various biological processes.

The combination of nucleosome positioning and dynamic modification of DNA and histones has a key role in gene regulation and guides development and differentiation.

Chromatine states can influence transcription directly by altering the packaging of DNA to allow or prevent access to DNA-binding proteins, or they can modifiy the nucleosome surface to enhance or impede recruitment of effector protein complexes.

This interplay between chromatin and transcription is dynamic and more complex than previously appreciated.

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ChIP-seq better than ChIP-chip

In ChIP-seq, the DNA fragments of interest are sequenced directly instead of being hybridized on an array.

ChIP-seq has higher resolution, fewer artefacts, greater coverage and a larger dynamic range than ChIP-chip, and therefore provides substantially improved data.

Although the short reads (~35bp) generated by NGS platforms pose serious difficulties for certain applications - for example, de novo genome assembly - they are aceptable for ChIP-seq.

The more precise mapping of protein-binding sites provided by ChIP-seq allows for a more accurate list of targets for transcription factors and enhancers, in addition to better identification of sequence motifs.

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ChIP profiles

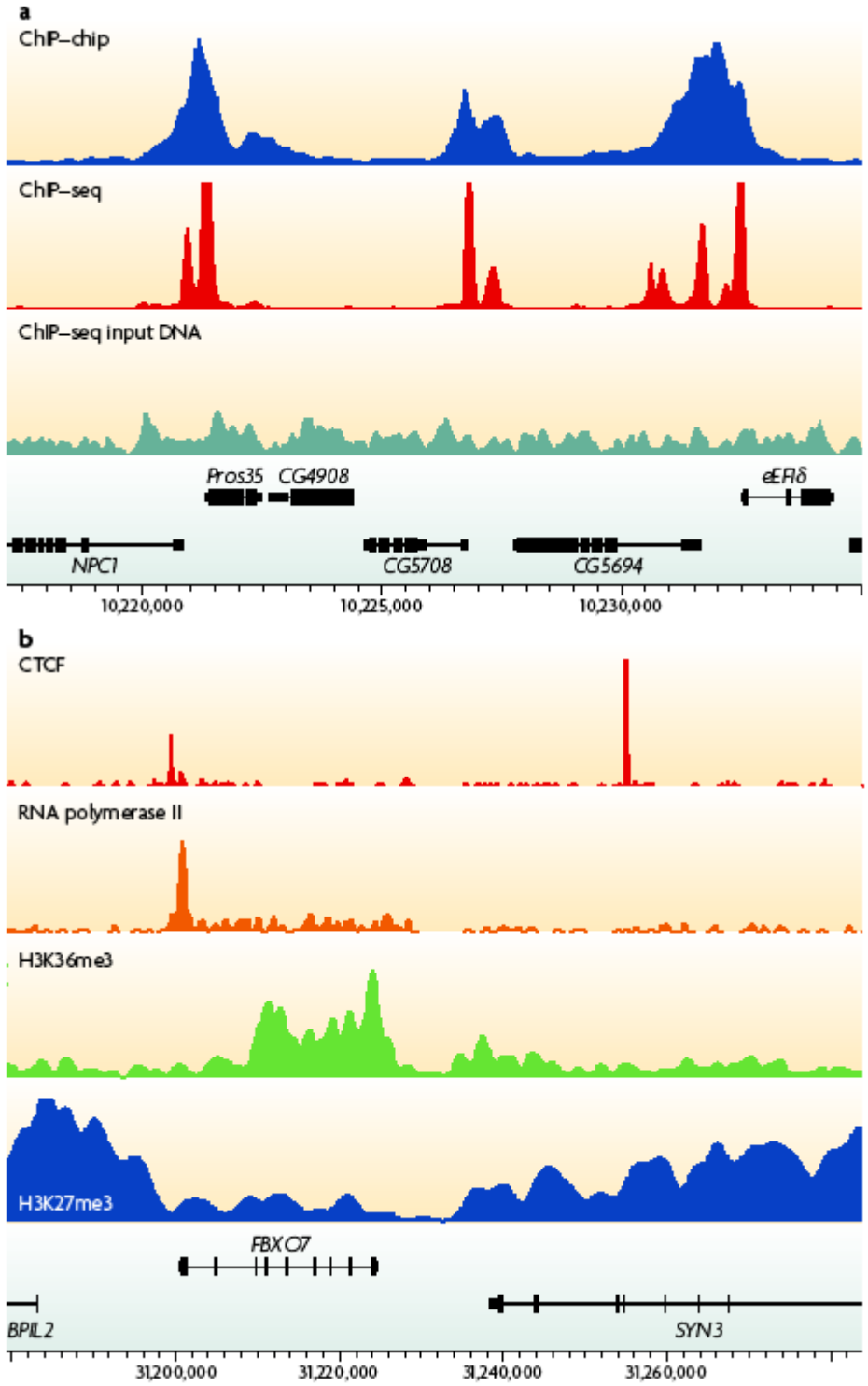

Figure 2 | **ChIP profiles. a** | Examples of the profiles generated by chromatin immunoprecipitation followed by sequencing (ChIP–seq) or by microarray (ChIP–chip). Shown is a section of the binding profiles of the chromodomain protein Chromator, as measured by ChIP–chip (unlogged intensity ratio; blue) and ChIP–seq (tag density; red) in the *Drosophila melanogaster* S2 cell line. The tag density profile obtained by ChIP–seq reveals specific positions of Chromator binding with higher spatial resolution and sensitivity. The ChIP–seq input DNA (control experiment) tag density is shown in grey for comparison. **b** | Examples of different types of ChIP–seq tag density profiles in human T cells. Profiles for different types of proteins and histone marks can have different types of features, such as: sharp binding sites, as shown for the insulator binding protein CTCF (CCCTC-binding factor; red); a mixture of shapes, as shown for RNA polymerase II (orange), which has a sharp peak followed by a broad region of enrichment; medium size broad peaks, as shown for histone H3 trimethylated at lysine 36 (H3K36me3; green), which is associated with transcription elongation over the gene; or large domains, as shown for histone H3 trimethylated at lysine 27 (H3K27me3; blue), which is a repressive mark that is indicative of Polycomb-mediated silencing. *BPIL2*, bactericidal/permeability-increasing protein-like 2; *FBXO7*, F box only 7; *NPC1*, Niemann-Pick disease, type C1; *Pros35*, proteasome 35 kDa subunit; *SYN3*, synapsin III. Data for part **b** are from REF. 25.

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Advantages of ChIP-seq over ChIP-chip

Table 1 | **Comparison of ChIP–chip and ChIP–seq**

	ChIP–chip	**ChIP–seq**
Maximum resolution	Array-specific, generally 30–100 bp	Single nucleotide
Coverage	Limited by sequences on the array; repetitive regions are usually masked out	Limited only by alignability of reads to the genome; increases with read length; many repetitive regions can be covered
Cost	US$400–800 per array (1–6 million probes); multiple arrays may be needed for large genomes	Currently US$1,000–2,000 per lane (using the Illumina Genome Analyzer); 6–15 million reads before alignment
Source of platform noise	Cross-hybridization between probes and nonspecific targets	Some GC bias can be present
Experimental design	Single- or double-channel, depending on the platform	Single channel
Cost-effective cases	Profiling of selected regions; when a large fraction of the genome is enriched for the modification or protein of interest (broad binding)	Large genomes; when a small fraction of the genome is enriched for the modification or protein of interest (sharp binding)
Required amount of ChIP DNA	High (a few micrograms)	Low (10–50 ng)
Dynamic range	Lower detection limit; saturation at high signal	Not limited
Amplification	More required	Less required; single-molecule sequencing without amplification is available
Multiplexing	Not possible	Possible

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Drawbacks of ChIP-seq

All profiling technologies produce unwanted artefacts, and ChIP-seq is no exception. Although sequencing errors have been reduced substantially as the technology has improved, they are still present, especially towards the end of each read.

There is also bias towards GC-rich content in fragment selection, both in library preparation and in amplification before and during sequencing, although notable improvements have been made recently.

In addition, when an insufficient number of reads is generated, there is a loss of sensitivity or specificity in detection of enriched regions.

There are also technical issues in performing the experiment, such as loading the correct amount of sample: too little sample will result in too few tags.

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ChIP-Seq: technical considerations for obtaining high-quality data

COMMENTARY

ChIP-Seq: technical considerations for obtaining high-quality data

Benjamin L Kidder, Gangqing Hu & Keji Zhao

Chromatin immunoprecipitation followed by next-generation sequencing analysis (ChIP-Seq) is a powerful method with which to investigate the genome-wide distribution of chromatin-binding proteins and histone modifications in any genome with a known sequence. The application of this technique to a variety of developmental and differentiation systems has provided global views of the *cis*-regulatory elements, transcription factor function and epigenetic processes involved in the control of gene transcription. Here we describe several technical aspects of the ChIP-Seq assay that diminish bias and background noise and allow the consistent generation of high-quality data.

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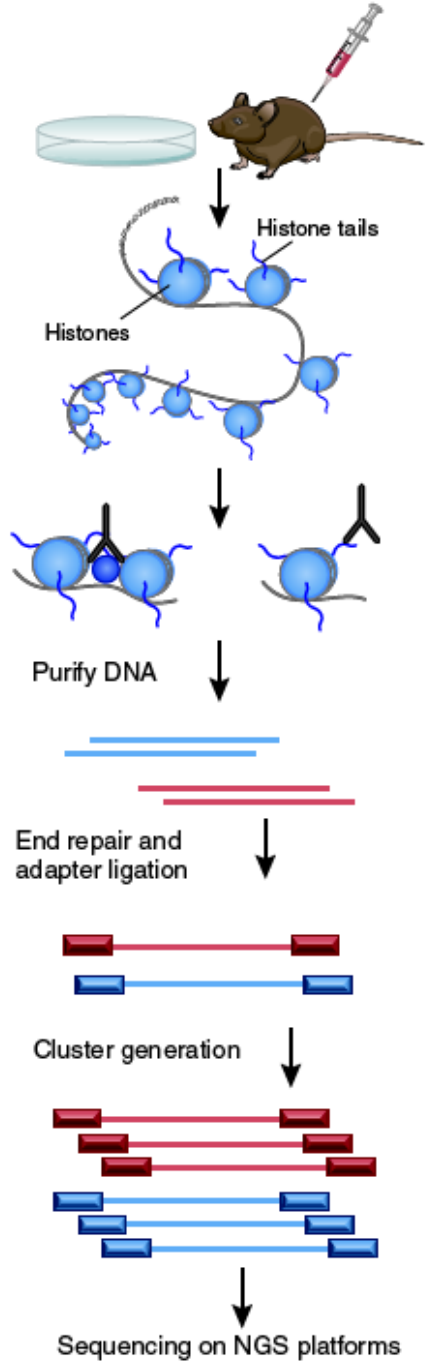
Immune function involves the tight control of transcription by the interplay of *cis*-regulatory elements and *trans*-acting transcription factors; this is influenced mainly by the epigenetic landscape of cells of the immune response. Published studies have identified many transcription factors and chromatin modifiers linked to innate and adaptive immunity. However, traditionally only one or a few individual gene(s) or regulatory region(s) have been studied by techniques such as chromatin immunoprecipitation (ChIP)-PCR assays. Technical advances in global techniques such and to elucidate immune memory[1–4]. Here we highlight several technical aspects of ChIP-Seq that should be considered for the generation of high-quality genome-wide data, including antibodies, controls, library construction and statistical analysis.

Antibodies

The quality of antibodies used for ChIP-Seq is one of the most important factors that contribute to the quality of the data generated by these studies. Antibodies that offer high sensitivity and specificity are necessary for ChIP-Seq related family members that may serve unique or redundant roles in the immune system. The specificity of an antibody can be directly addressed by immunoblot analysis of a protein of interest through the use of models of knockdown by RNA-mediated interference or genetic knockout. In these cases, because expression of the protein should be diminished to background amounts, any protein detected by immunoblot analysis can be assumed to be nonspecific. ChIP with a higher concentration of antibody, which can be acquired from various companies by request, or pooled mono-

Nat Immunol. 2011 Sep 20;12(10):918-22

ChIP-Seq: technical considerations for obtaining high-quality data



Isolation of cells of the immune response

- Use the correct number of cells: 1×10^6 to 10×10^6
- Collect biological replicates of cells
- Choose an appropriate control for antibody specificity (knockout or RNAi knockdown)

Fragmentation by sonication or MNase treatment

- Shear chromatin to a size range of ~150–300 bp
- Sonicate chromatin extracts for non-histone proteins
  - * Sonication conditions should be determined empirically for each cell type
- Treat chromatin extracts with MNase for analysis of histone modifications
  - * Do not overdigest chromatin

ChIP analysis of histone modifications, transcription factors or epigenetic regulators

- Select antibody: monoclonal versus polyclonal
- Choose reference control (Input or IgG)
- Perform ChIP with established protocols
- Purify DNA

Library construction

- Do end repair and adapter ligation
- Perform PCR using primers compatible with sequencing platform
  - * Avoid overamplifying DNA

Sequencing

- Determine sequencing depth on the basis of the prevalence of binding throughout the genome: more sequencing tags may be needed for diffuse signals (such as H3K27me3)
- Perform single-end or paired-end sequencing

Figure 1 ChIP-Seq experimental design. ChIP-Seq is a powerful tool with which to investigate protein-DNA interactions on a global scale. It is important that the appropriate controls for antibody specificity be determined before ChIP-Seq is begun. After isolation of the ideal number of cells, chromatin is sheared into an ideal size range by sonication or enzymatic means (micrococcal nuclease (MNase)). Next, high-quality antibodies are used for ChIP to enrich for factor-occupied DNA sequences. After purification of ChIP-enriched DNA, a library is constructed to allow sequencing on next-generation sequencing (NGS) platforms. Library construction typically includes end-repair, the addition of single adenosine residues, adaptor ligation and PCR with primers compatible with the sequencing platform. After cluster generation, single- or paired-end sequencing is performed on next-generation sequencing platforms. RNAi, RNA-mediated interference; bp, base pairs.

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ChIP-seq: technical considerations for obtaining high-quality data

There are several technical aspects of ChIP-seq that should be considered for the generation of high quality genome wide data, including antibodies, controls, library construction and statistical analysis.

Antibody quality
ChIP-seq data depends crucially on the quality of the antibody used.

A sensitive and specific antibody will give a high level of enrichment compared with the background, which makes it easier to detect binding events.

Many antibodies are commercially available, and some are noted as ChIP grade, but the quality of different antibodies is highly variable and can also vary among batches of a specific antibody.

Rigorous validation is a laborious process: for histone modifications for instance, the reactivity of the antibody with unmodified histones or non-histone proteins should be checked by western blotting.

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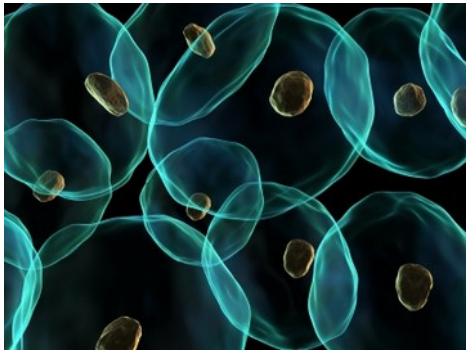
ChIP-Seq: technical considerations for obtaining high-quality data



Cell number

The abundance of the protein or histone modification to be investigated and the quality of the antibody should be considered when determining the number of cells to begin ChIP-seq analysis.

As the signal-to-noise ratio is directly correlated with the cell number, the use of more cells tends to produce a higher signal-to-noise ratio. Therefore, it is important to empirically determine the minimum number of cells that can be used, whenever possible.

ChIP-seq experiments typically require $1x10^6$ to $10x10^6$ cells, which results in 10-100 ng of ChIP DNA. The former ($1x10^6$ cells) is usually sufficient for the analysis of abundant proteins such as RNA polymerase II and localized histone modifications such as trimethylation of histone H3 and Lys4 (H3K4me3), whereas the latter ($10x10^6$) may be required for the analysis of less-abundant proteins or difuse histone modifications.

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ChIP-Seq: technical considerations for obtaining high-quality data

Chromatin fragmentation
Before ChIP, chromatin must be fragmented into a manageable size (~ 150-300 base pairs) by sonication or enzymatic means (usually via treatment with micrococcal nuclease).

For histone modification, digestion of native chromatin into mononucleosome-sized particles with micrococcal nuclease may be the preferred method, because this generates high-resolution data for nucleosome modifications and eliminates signal artifacts caused by crosslinking with other genomic regions.

For mapping of binding sites for transcription factors, sonication of formaldehyde-crosslinked chromatin is the more appropriated method.

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ChIP-Seq: technical considerations for obtaining high-quality data

Chromatin fragmentation

The conditions used to sonicate chromatin need to be optimized for each cell type because they are highly variable and depend on the cell type, the number of cells used, fixation conditions, type of sonicator and sonicator settings.

It is important to avoid oversonication of chromatin when transcription factors are to be evaluated by ChIP, whereas oversonication may not be as problematic for analysis of histone modifications.

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ChIP-Seq: technical considerations for obtaining high-quality data

Control experiment

The experimental steps in ChIP involve several potential sources of artifacts.

Shearing of DNA, for example, does not result in uniform fragmentation of the genome: open chromatin regions tend to be fragmented more easily than closed regions, which creates an uneven distribution of sequence tags across the genome.

Also, repetitive sequences might seem to be enriched because of inaccuracies in the number of copies of the repeats in the assembled genome.

Therefore, a peak in the ChIP-seq profile should be compared with the same region in a matched control sample to determine its significance.

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Control experiment

There are different types of control sample (negative controls used to substrat the peak signal):

a) **Input DNA** (a portion of the DNA sample removed prior to immunoprecipitation (IP))

b) **DNA from nonspecific IP** (IP performed using an antibody, such as nonspecific immunoglobulin G).

The IP for the factor of interest at a positive control site (where it is expected to bind at high level) should be several fold above that of the control.

Input DNA has been used as the control sample in most ChIP-seq studies.

Most nonspecific IgG antibodies usually immunoprecipitate much less DNA than specific antibodies, thus the resulting control reads will not cover the genome as sufficiently as a background model would for peak identification.

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ChIP-Seq: technical considerations for obtaining high-quality data

Replicates
Many factors, including cell-culture conditions, ChIP and library construction, may contribute to variability between data sets.

To ensure reliability of the data, biological replicate experiments are necessary.

Although there is no consensus on the correct number of replicates needed, at least duplicate biological experiments should be done.

Although only one ChIP-grade antibody is available for the analysis of most histone modifications and transcription factors, it is recommended that ChIP-seq data be confirmed through the use of a different antibody wherever possible, to control for a potential antibody cross-reactivity.

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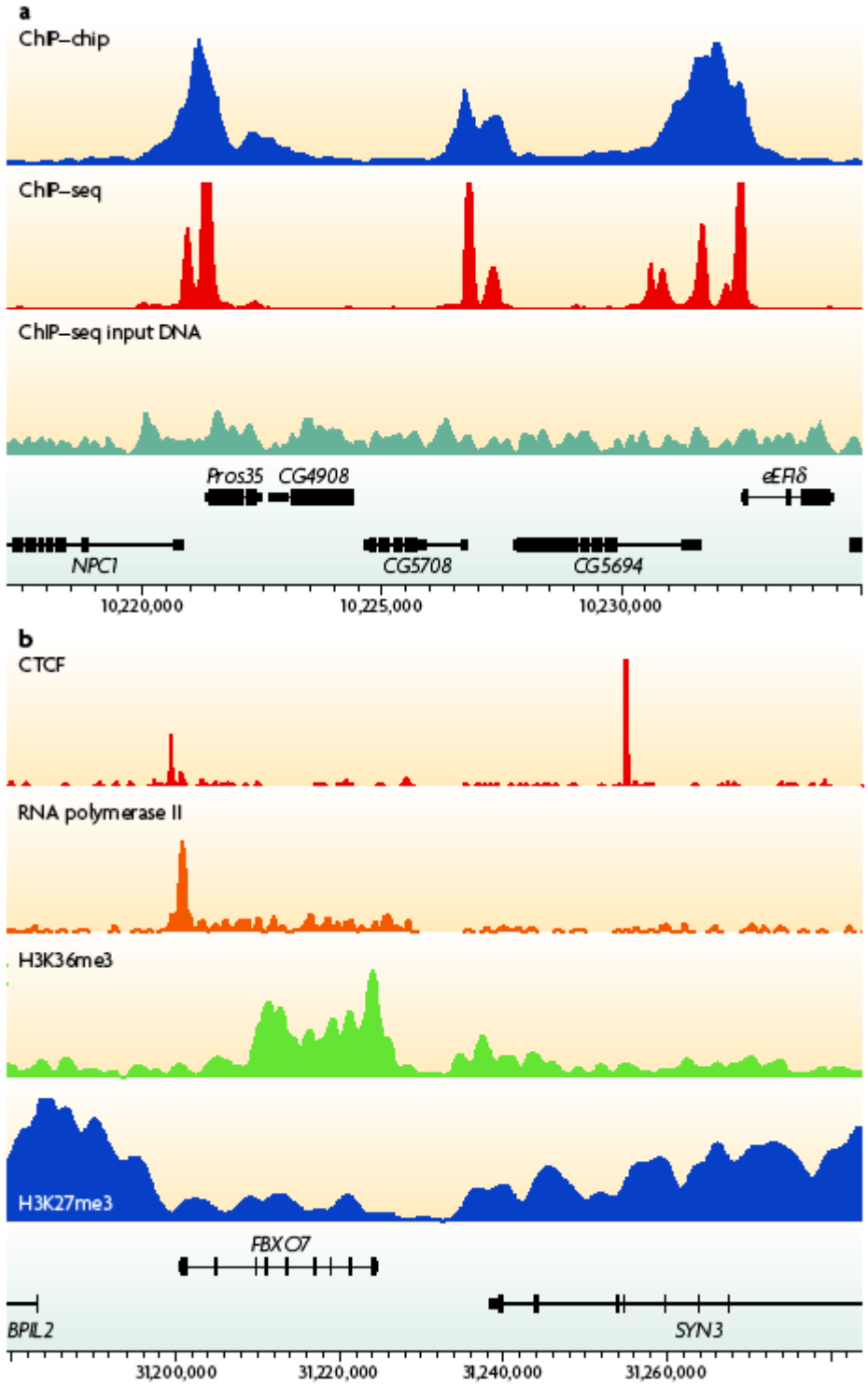
ChIP profiles



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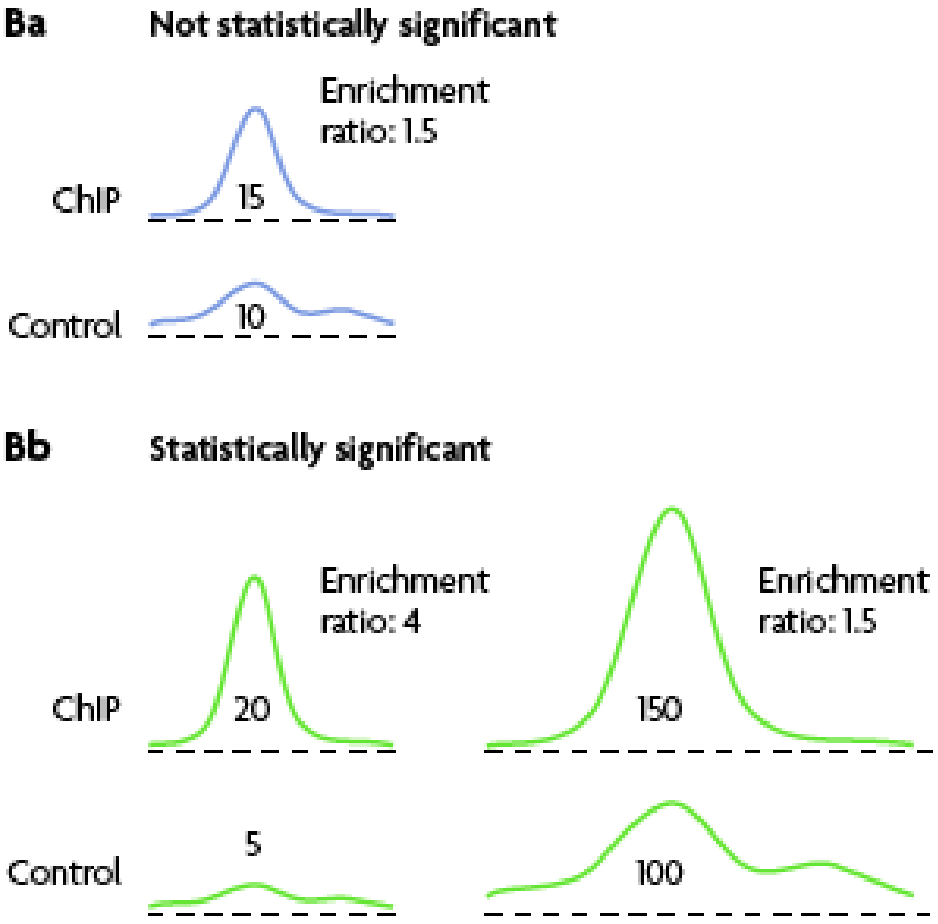
Depth of sequencing

Ba. A peak that is not statistically significant - the enrichment ratio between the ChIP and control experiments is low (1.5).

Bb. Two ways in which a peak can be statistically significant.
On the left, although the number of tag counts is low, the enrichment ratio between the ChIP and control experiments in high (4).
On the right, the peaks have the same ratio as those in Ba, but have a large number of tag counts.



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Identification of enriched regions

After sequenced reads are aligned to the genome, the next step is to identify regions that are enriched in the ChIP sample relative to the control with statistical significance.

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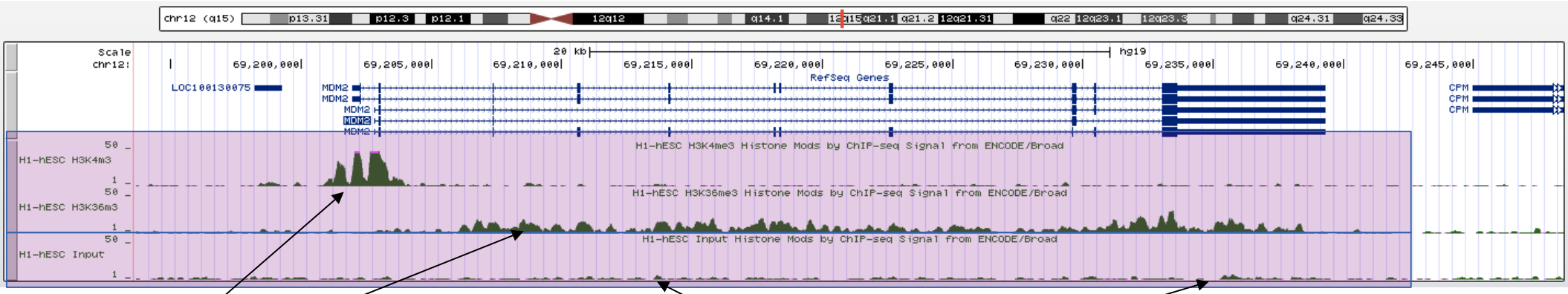
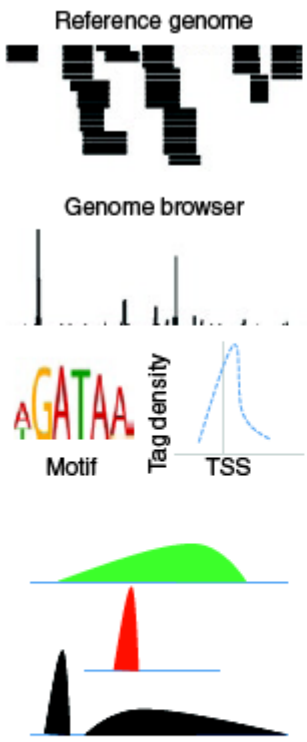


ChIP-seq peaks

Control sample signal

ChIP-Seq: Data analysis



Sequence alignment

(no longer a bottleneck for data analysis)

Inspect data quality

- Check report from vendor-supplied analytical pipeline
- Visually inspect with a genome browser
- Identify motif in tag-enriched region
- Examine the profile at certain genomic features
- Confirm by quantitative PCR

Peak calling (choose the right tool)

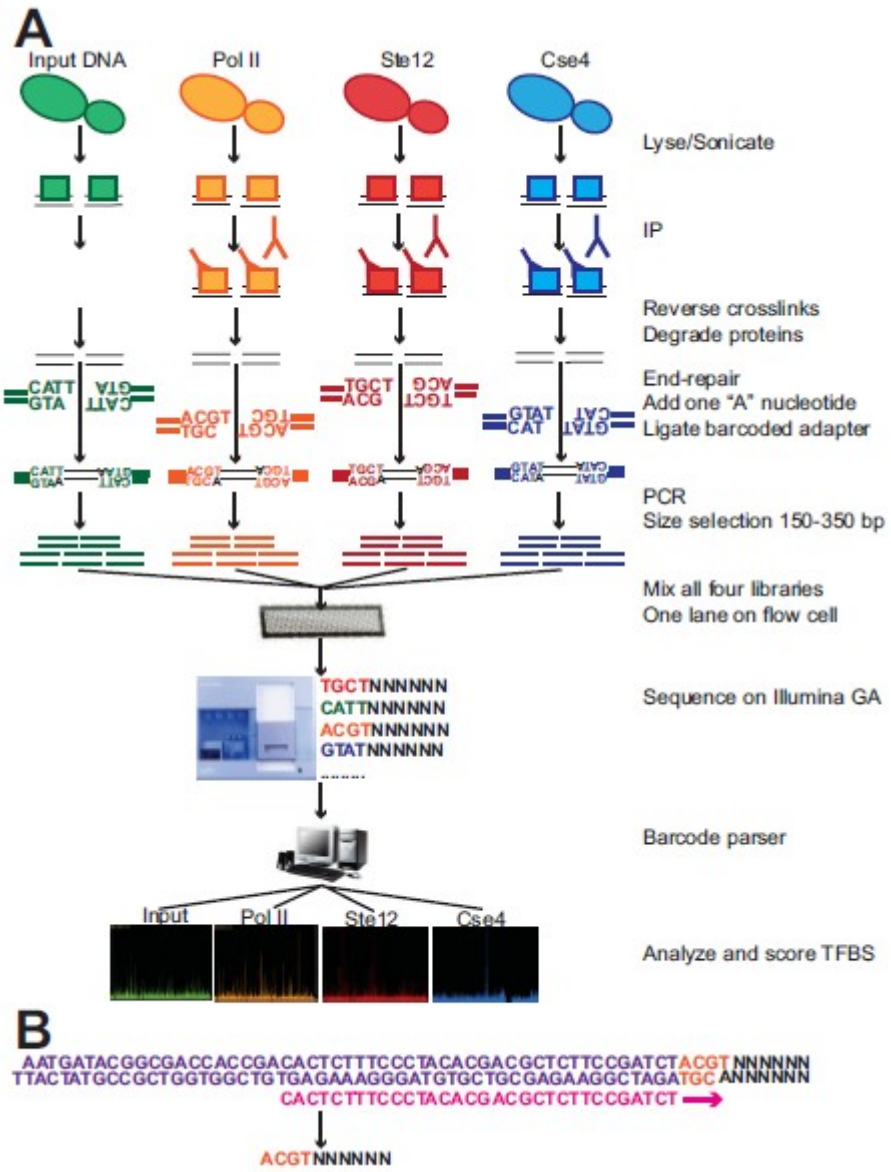
Type of peak	Example	Representative tools
Broad	H3K27me3	CCAT, SICER
Sharp	CTCF	MACS
Sharp & broad	Pol II	ZINBA

Figure 2 Common procedures for ChIP-seq data analysis. After base-calling, short-read sequences are aligned to a reference genome. Data quality is assessed by a combination of various strategies, such as visual inspection with a genome browser, motif identification and confirmation by ChIP and quantitative PCR. The initial inspection or prior knowledge provides information about whether the peaks are broad or sharp or both. Various algorithms (bottom right) have been developed for the identification of peaks of these three groups.

Nat Immunol. 2011 Sep 20;12(10):918-22

Multiplexing

Scheme for yeast barcoded ChIP-Seq. (a) Barcoded ChIP-Seq workflow. Ovals depict yeast cells and squares depict proteins. An aliquot of sheared cell lysate is not immunoprecipitated but is otherwise processed normally (green). This DNA, termed input DNA, is a reference sample for ChIP-Seq. Illumina DNA libraries are generated from both ChIP and input DNA samples. In multiplex ChIP-Seq, a barcoded adapter is ligated to an individual DNA sample. The barcode has 3 unique bases followed by a 'T' to anneal with the end-repaired DNA. Four libraries are then pooled together and applied to a single flowcell lane. After sequencing on the Genome Analyzer, reads are separated according to the first four bases and aligned to the yeast genome. Reads are stacked to generate a signal profile and scored against a pool of input DNA to determine significant transcription factor binding sites. (b) The barcode (orange) is located between Illumina adapter sequences (purple) and ChIP or input DNA inserts (black). The sequencing primer (pink) anneals to the adapter sequences and short reads start with the four bases of the barcode (orange) followed by DNA inserts (black). For the sequencing primer and Illumina adapter, oligonucleotide sequences were given by the manufacturer © 2006 Illumina, Inc. All rights reserved.



BMC Genomics. 2009 Jan 21;10:37.

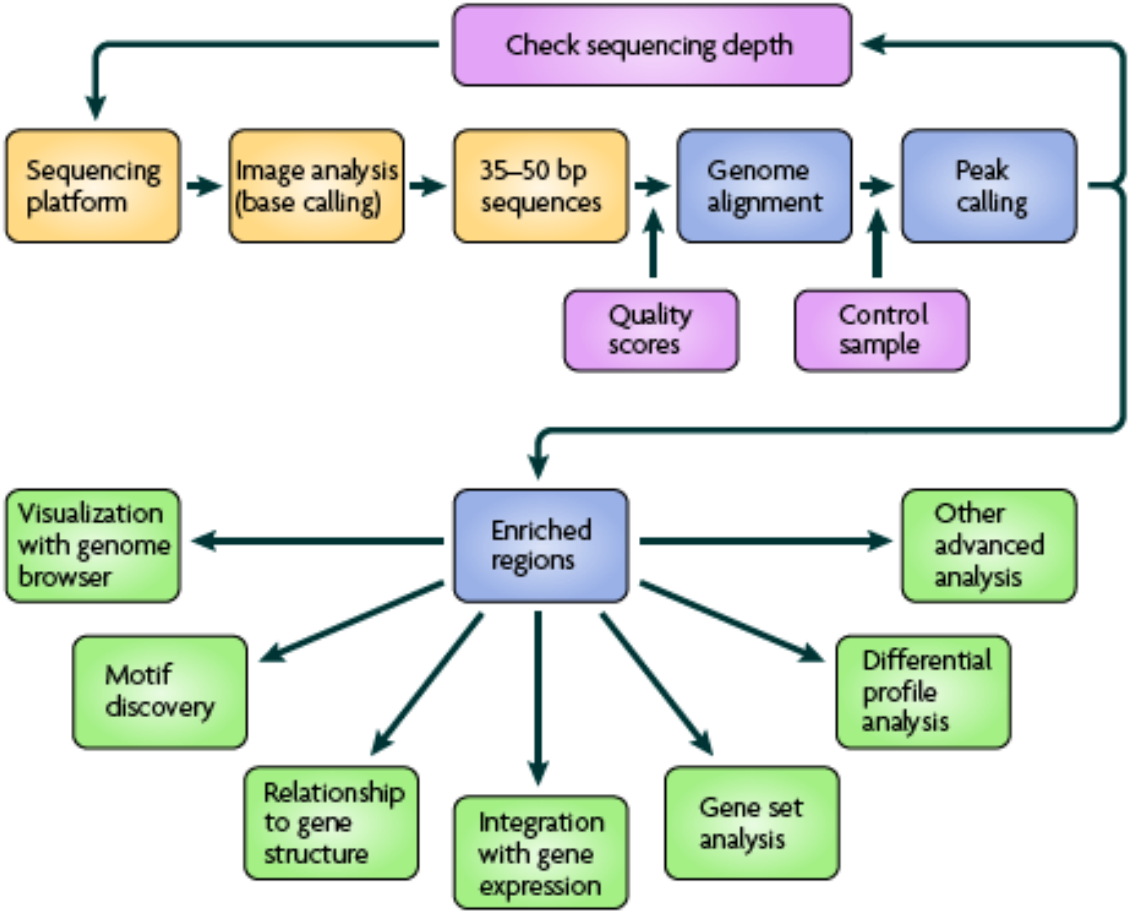


Figure 4 | **Overview of ChIP–seq analysis.** The raw data for chromatin immunoprecipitation followed by sequencing (ChIP–seq) analysis are images from the next-generation sequencing platform (top left). A base caller converts the image data to sequence tags, which are then aligned to the genome. On some platforms, they are aligned with the aid of quality scores that indicate the reliability of each base call. Peak calling, using data from the ChIP profile and a control profile (which is usually created from input DNA), generates a list of enriched regions that are ordered by false discovery rate as a statistical measure. Subsequently, the profiles of enriched regions are viewed with a browser and various advanced analyses are performed.

Nat Rev Genet. 2009 Oct;10(10):669-80.

Challenges in data analysis

very important:

Validation of a number of peaks is always recommended in a ChIP-seq analysis !!!

Nat Rev Genet. 2009 Oct;10(10):669-80.

ENCODE recomendations for ChIP-seq

Resource

ChIP-seq guidelines and practices of the ENCODE and modENCODE consortia

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Chromatin immunoprecipitation (ChIP) followed by high-throughput DNA sequencing (ChIP-seq) has become a valuable and widely used approach for mapping the genomic location of transcription-factor binding and histone modifications in living cells. Despite its widespread use, there are considerable differences in how these experiments are conducted, how the results are scored and evaluated for quality, and how the data and metadata are archived for public use. These practices affect the quality and utility of any global ChIP experiment. Through our experience in performing ChIP-seq experiments, the ENCODE and modENCODE consortia have developed a set of working standards and guidelines for ChIP experiments that are updated routinely. The current guidelines address antibody validation, experimental replication, sequencing depth, data and metadata reporting, and data quality assessment. We discuss how ChIP quality, assessed in these ways, affects different uses of ChIP-seq data. All data sets used in the analysis have been deposited for public viewing and downloading at the ENCODE (http://encodeproject.org/ENCODE/) and modENCODE (http://www.modencode.org/) portals.

[Supplemental material is available for this article.]

Methods for mapping transcription-factor occupancy across the genome by chromatin immunoprecipitation (ChIP) were developed more than a decade ago (Ren et al. 2000; Iyer et al. 2001; Lieb et al. 2001; Horak and Snyder 2002; Weinmann et al. 2002). In ChIP assays, a transcription factor, cofactor, or other chromatin protein of interest is enriched by immunoprecipitation from cross-linked cells, along with its associated DNA. Genomic DNA sites enriched in this manner were initially identified by DNA hybridization to a microarray (ChIP-chip) (Ren et al. 2000; Iyer et al. 2001; Lieb et al. 2001; Horak and Snyder 2002; Weinmann et al. 2002), and more recently by DNA sequencing (ChIP-seq) (Barski et al. 2007; Johnson et al. 2007; Robertson et al. 2007). ChIP-seq has now been widely used for many transcription factors, histone modifications, chromatin modifying complexes, and other chromatin-associated proteins in a wide variety of organisms. There is, however, much diversity in the way ChIP-seq experiments are designed, executed, scored, and reported. The resulting variability and data quality issues affect not only primary measurements, but also the ability to compare data from multiple studies or to perform integrative analyses across multiple data-types.

The ENCODE and modENCODE consortia have performed more than a thousand individual ChIP-seq experiments for more than 140 different factors and histone modifications in more than 100 cell types in four different organisms (*D. melanogaster*, *C. elegans*, mouse, and human), using multiple independent data production and processing pipelines (The ENCODE Project

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MACS (Model-based Analysis for ChIP-seq)

http://liulab.dfci.harvard.edu/MACS/

Open Access

Method

Model-based Analysis of ChIP-Seq (MACS)

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Abstract

We present Model-based Analysis of ChIP-Seq data, MACS, which analyzes data generated by short

Genome Biol (2008) vol. 9 (9) pp. R137

MACS (Model-based Analysis for ChIP-seq)

MACS performs peak-calling from ChIP-seq mapped reads through two main steps:

1.- Modeling the shift size of ChIP-seq tags
2.- Peak detection

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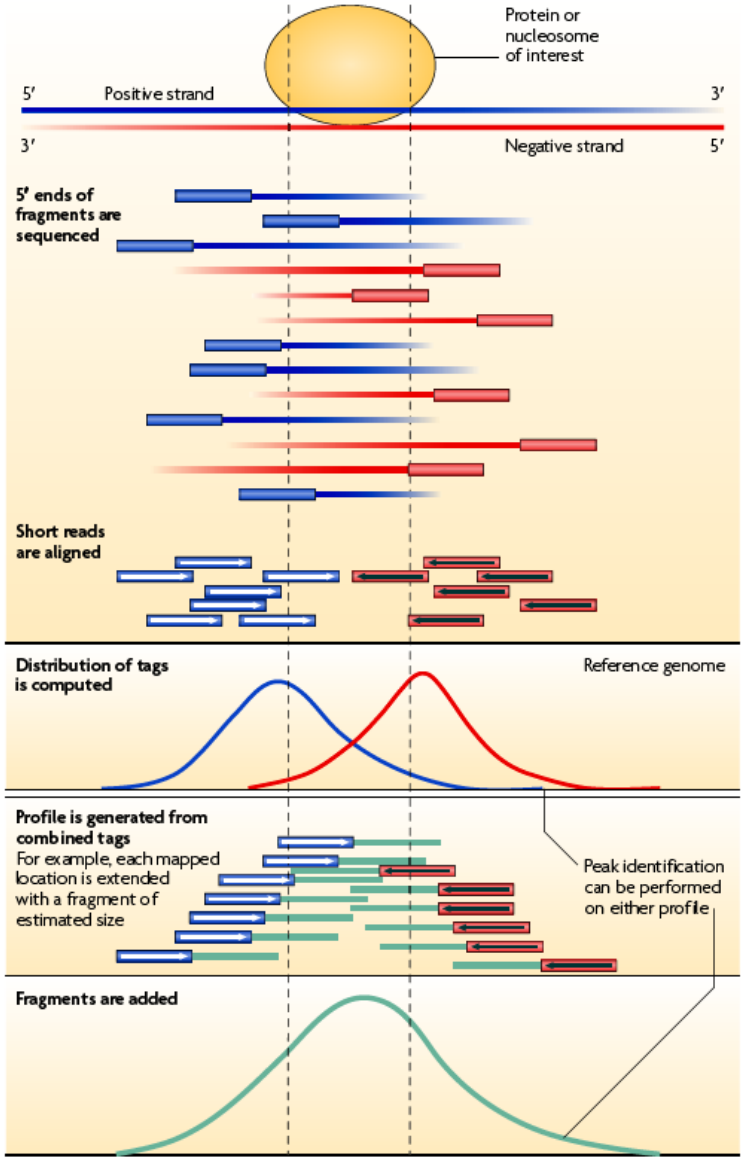
Challenges in data analysis

Identification of enriched regions

The fragments are sequenced at the 5' end and the locations of mapped reads should form two distributions, one on the positive strand and the other on the negative strand, with a consistent distance between the peaks of the distributions.

In these methods, a smoothed profile of each strand is constructed and the combined profile is calculated either by shifting each distribution towards the centre or by extending each mapped position.



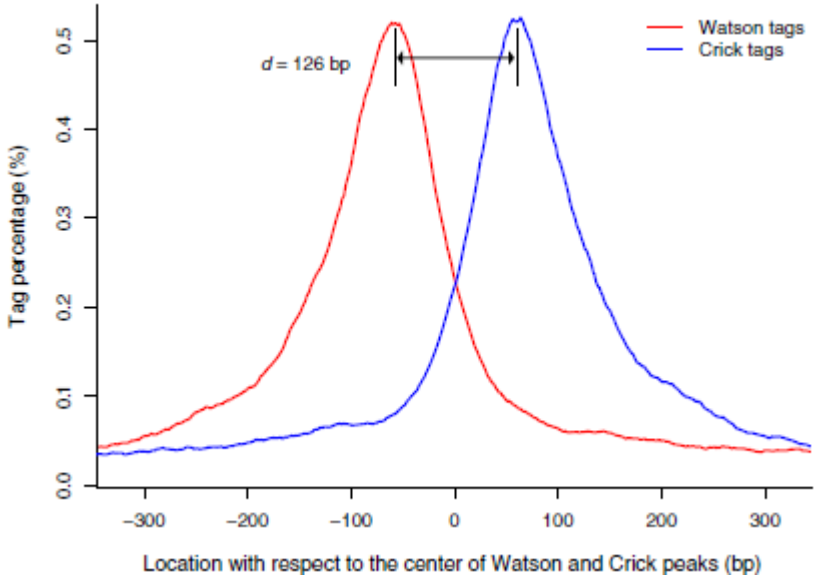
Strand-specific profiles at enriched sites

Nat Rev Genet. 2009 Oct;10(10):669-80.

MACS: Modeling the shift size of ChIP-seq

Given a sonication size (bandwith) and a high-confidence fold-enrichment (mfold), MACS slides 2xbandwith size windows across the genome to find regions with tags more than mfold enriched relative to a random tag genome distribution.

After that, MACS randomly samples 1000 of these high-quality peaks, separates their Watson and Crick tags, and aligns them between their Watson and Crick tag centers (requires shifting).



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MACS: Modeling the shift size of ChIP-seq

Why this approach is expected to work?

ChIP-seq tags represent the ends of fragments in a ChIP-DNA library.

Since ChIP-DNA fragments are equally likely to be sequenced from both ends, the tag density around a true binding site should show a bimodal enrichment pattern, with Watson strand tags enriched upstream of binding and Crick tags enriched downstream.

MACS takes advantage of this bimodal pattern to empirically model the shifting size to better locate the real binding sites.

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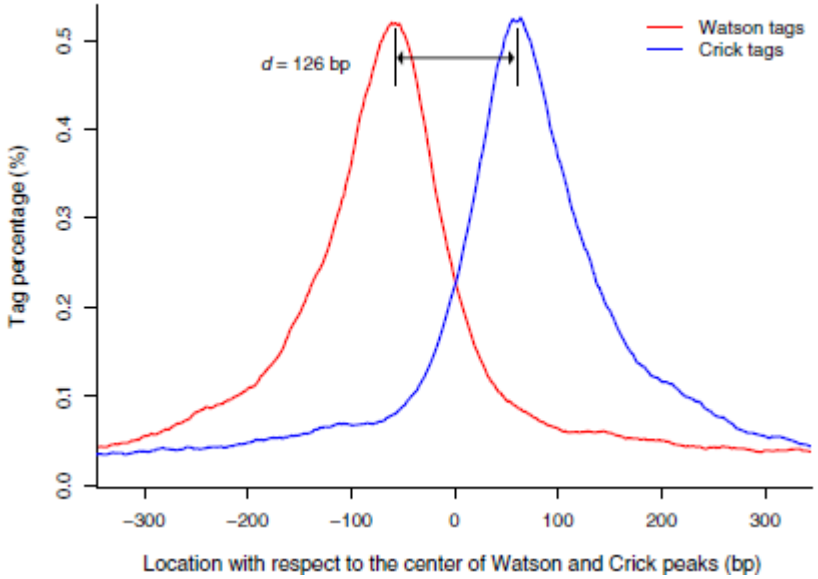
MACS: Modeling the shift size of ChIP-seq

The distance between the modes of the Watson and Crick peaks in the alignment is defined as 'd' and MACS shifts all the tags by d/2 toward the 3' ends to the most likely protein-DNA interaction sites.



Modelling of the shift size is a way for MACS to guide peak detection.

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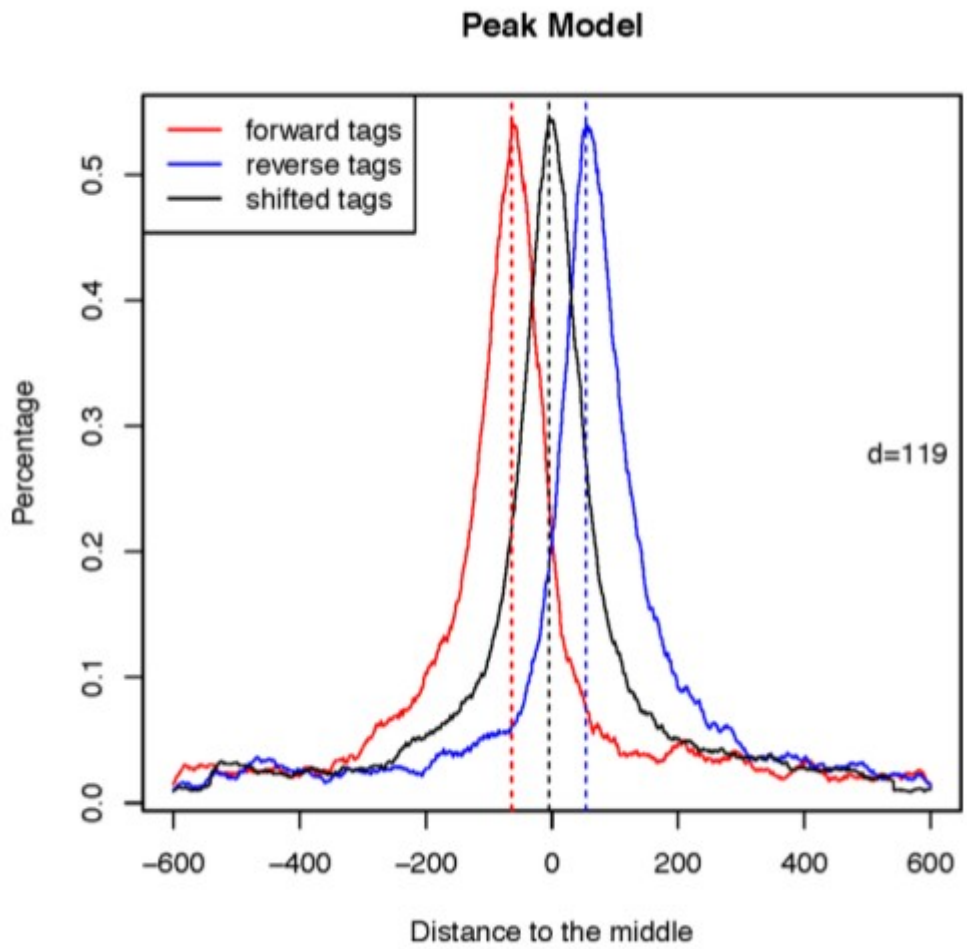
MACS: Modeling the shift size of ChIP-seq



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MACS: FDR estimation for detected peaks

For a ChIP-seq experiment with controls, MACS empirically estimates the false discovery rate (FDR) for each detected peak.

MACS uses the same parameters to find ChIP peaks over control and control peaks over ChIP (that is, **a sample swap**).

The empirical FDR is defined as 'number of control peaks / number of ChIP peaks' at each p-value.

MACS can also be applied to differential binding between two conditions by treating one of the samples as the control.

****Since peaks from either sample are likely to be biologically meaningful in this case, we cannot use a sample swap to calculate FDR, and the data quality of each sample needs to be evaluated against a real control.

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MACS: Discussion

ChIP-seq users are often curious as to whether they have sequenced enough to saturate all the binding sites. In principle, sequencing saturation should be dependent on the fold-enrichment, since higher-fold peaks are saturated earlier than lower-fold ones.

MACS produces a saturation table to report, at different fold-enrichments, the proportion of sites that could still be detected when using 90% to 20% of the tags.

They saw that while peaks with over 60-fold enrichment have been saturated, deeper sequencing could still recover more sites less than 40-fold enriched relative to the chromatine input DNA.

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MACS: Recommendations

When tag counts from ChIP and controls are not balanced, the sample with more tags often gives more peaks even though MACS normalizes the total tag counts between the two samples.

They suggest to ChIP-seq users that if they sequence more ChIP tags than controls, the FDR estimate of their ChIP peaks might be overly optimistic.

If a user has replicated files for ChIP or control, it is recommended to concatenante all replicates into one input file: pool of replicates.

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Thanks for your attention !!